

# L1 Regularisation Values for Inverse-Design

## Surrogate Optimisation

Extracted: 2026-06-20

### Extracted lambda\_L1 Values

The table below lists the L1 regularisation strength extracted from every JSON file in runtime\_data/surrogate\_optimisation/. No files were missing and no lambda\_L1 key was absent from any file.

| Target state | Node number | File        | lambda_L1 |
|--------------|-------------|-------------|-----------|
| GHZ          | 4           | GHZ/4n.json | 1e-4      |
| GHZ          | 6           | GHZ/6n.json | 1e-5      |
| GHZ          | 8           | GHZ/8n.json | 1e-6      |
| W            | 4           | W/4n.json   | 1e-4      |
| W            | 6           | W/6n.json   | 1e-5      |
| W            | 8           | W/8n.json   | 1e-6      |
| LC           | 4           | LC/4n.json  | 1e-4      |
| LC           | 6           | LC/6n.json  | 1e-5      |
| LC           | 8           | LC/8n.json  | 1e-6      |

### Pattern

The regularisation strength is identical across all three target states (GHZ, W, LC) and decreases by one order of magnitude with each two-node increase in system size:

- 4-node:  $\lambda = 1e-4$
- 6-node:  $\lambda = 1e-5$
- 8-node:  $\lambda = 1e-6$

### Suggested Methods Sentence

*"The L1 regularisation strength was set to  $\lambda = 1e-4$ ,  $1e-5$ , and  $1e-6$  for the 4-, 6-, and 8-node systems, respectively, applied uniformly across all target states (GHZ, W, and LC)."*